

D E C K 1 7 · W E E K 7 · M O R N I N G 1

Derivatives I

Forwards, Futures & Swaps

Firm-commitment instruments and their no-arbitrage pricing

CFA Level I · Vol 7 Learning Modules 1–7

Roadmap — Derivatives I: Forwards, Futures & Swaps

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Derivative markets & instruments

Definition · OTC vs ETD · ISDA · central clearing · firm commitments vs contingent claims

02

Benefits, risks & uses

Risk transfer · price discovery · ops efficiency · six curriculum risks · issuer vs investor

03

Arbitrage, replication & cost of carry

Law of one price · cash-and-carry arbitrage · replication identity · dividends / storage / income adjustments

04

Forwards & futures

$F_0(T) = S_0(1+r)^T$ · MTM over life · equity / FX / index forwards · IRP · daily margining · basis · convexity bias

05

Swaps

Strip of forwards · par swap rate · $PV(\text{floating}) = PV(\text{fixed})$ · IRS = long FRN + short fixed bond · valuation over time

Learning objectives

By the end of this deck you should be able to:

1. Describe the basic features of a derivative and contrast OTC vs exchange-traded markets, including the role of ISDA, central clearing and margining.
2. Distinguish firm commitments (forwards, futures, swaps) from contingent claims (options, credit derivatives) by their payoff profile and settlement mechanics.
3. Evaluate the benefits (risk transfer, information discovery, operational advantages, market efficiency) and the six curriculum risks of derivative use.
4. Apply the law of one price to price a forward via cash-and-carry arbitrage and adjust the formula for income, storage costs, and interest-rate differentials.
5. Price and value forward contracts on equities (with discrete dividends and dividend yields), FX (via covered interest-rate parity), and interest rates (FRAs).
6. Price and value plain-vanilla interest-rate swaps by solving for the par swap rate and by interpreting the swap as a pair of offsetting bond positions.

SECTION 01

Derivative markets & instruments

A derivative is a contract whose value is derived from an underlying — the buyer gets a long-like exposure, the seller a short-like exposure, at a specified maturity

A forward on 1,000 Airbus shares at EUR 30/share settles either in physical delivery or in cash for the spot-vs-forward difference.

$$\begin{aligned}\text{Payoff at } T \text{ (long)} &= S_T - F_0(T) \\ \text{Payoff at } T \text{ (short)} &= F_0(T) - S_T\end{aligned}$$

where: S_T = underlying spot price at maturity; $F_0(T)$ = forward price agreed at inception ($t = 0$). The long BUYS and gains when S_T rises.

UNDERLYING

Equities, fixed income (bonds, interest rates), FX, commodities, credit, volatility.
May be a basket, an index or a variable
— not necessarily a single asset.

COUNTERPARTIES

Legal entities with matching obligations.
Almost always one end-user (issuer or investor) and a financial intermediary.
Counterparty credit risk is a first-order concern.

SIZE & MATURITY

Notional (or contract size) agreed at outset. Maturity = time to settlement.
Cash settlement = net difference;
physical settlement = exchange of asset for cash.

Source: CFA Institute, Curriculum Vol 7, Learning Module 1, "Derivative Features", pp. 5–8 (printed).

OTC derivatives are customized bilateral contracts under an ISDA; exchange-traded derivatives are standardized, centrally cleared, and margined daily

The post-2008 central clearing mandate pushed most standardized OTC rate and FX swaps onto CCPs, narrowing the structural gap with ETDs.

Feature	Over-the-counter (OTC)	Exchange-traded (ETD)
Contract terms	Customized: size, maturity, underlying	Standardized contract spec
Counterparty credit risk	Bilateral – mitigated by ISDA + collateral	Borne by the CCP (novation)
Settlement	Typically at maturity	Daily mark-to-market via margin
Transparency / liquidity	Lower; dealer quotes, negotiated	Higher; public order book
Typical products	FX forwards, IRS, credit default swaps	Futures, listed options
Regulation	Clearing mandate if standardized	Exchange + national regulator

CENTRAL CLEARING BY NOVATION

A trade executed on a swap execution facility is submitted to a central counterparty, which NOVATES it — replacing one bilateral contract with two identical contracts facing the CCP. Bilateral credit risk disappears; it is concentrated in the CCP instead, which is why CCP safeguards are themselves a systemic question.

¹ Master Agreement, CSA, Dodd-Frank and EMIR are market context: Vol 7 names only the "ISDA Agreement" and a post-2008 central clearing mandate.

Source: CFA Institute, Curriculum Vol 7, Learning Module 1, "Derivative Markets", pp. 14–18 (printed).

Firm commitments fix the terms today and settle in the future; contingent claims give the buyer an optional payoff — forwards, futures and swaps vs options

Firm commitments have SYMMETRIC payoffs (buyer's gain = seller's loss). Contingent claims have ASYMMETRIC payoffs (buyer caps loss at premium).

FIRM COMMITMENTS

Both parties are OBLIGED to transact at the pre-agreed terms.

- Forward — OTC, one-shot, settled at maturity
- Future — ETD, one-shot at maturity but daily MTM in cash
- Swap — OTC, periodic exchange (usually IRS fixed vs MRR)

$V_0(T) = 0$ at inception (no cash changes hands).

Payoff is SYMMETRIC: $V_T(\text{long}) = S_T - F_0(T)$.

CONTINGENT CLAIMS

ONE party (the buyer) decides whether and when to exercise.

- Call option — right to buy at X
- Put option — right to sell at X
- Credit derivative (CDS) — payoff contingent on a credit event

Buyer pays a PREMIUM (c_0 or p_0) at inception $\rightarrow V_0 \neq 0$.

Payoff is ASYMMETRIC: buyer's loss is capped at premium.

Forwards, futures and swaps share a symmetric payoff profile and zero value at inception — they differ in settlement mechanics and in the cash-flow shape

Same underlying economic exposure to the forward curve — very different daily cash-flow and credit-risk profiles.

Feature	Forward	Futures	Swap
Market	OTC	Exchange-traded	OTC (cleared if std)
Contract	Customized	Standardized	Customized
Cash flow profile	Single exchange at T	Single exchange + daily MTM	Periodic over life
Credit risk	Counterparty (bilateral)	CCP (very low)	Counterparty + margin
Initial value	$V_0 = 0$	$V_0 = 0$ (resets daily)	$V_0 = 0$
Payoff shape	Symmetric	Symmetric	Symmetric

MENTAL MODEL

A swap is economically a STRIP of forwards, but priced at a SINGLE fixed rate (the par swap rate) rather than at each period's own forward rate. A futures contract is a forward wrapped in a daily settlement mechanism that eliminates most counterparty credit risk.

Source: CFA Institute, Curriculum Vol 7, Learning Module 2, "Forwards, Futures, and Swaps", pp. 26–37 (printed).

SECTION 02

Benefits, risks & uses

Derivatives deliver four core benefits – risk transfer, information discovery, operational advantages, and market efficiency – that cash markets alone cannot match

Same underlying exposure as the cash market, a fraction of the capital, and a bridge between an economic decision and the moment it can be transacted.

Benefit (CFA Exhibit 1)	Mechanism	Where LM 3 shows it
Risk allocation / transfer	Bridges decision-to-transaction timing gap	Montau KRW/EUR forward
Information discovery	Futures and implied vol reveal expectations	Ex 3: WTI at -USD 37.63
Operational advantages	Low outlay, low cost, liquid, easy to short	Ex 4: Procam gold margin
Market efficiency	Cheaper to arbitrage mispricings away	Printed pp. 61-62

CFA LM 3 EXAMPLE 4 – PROCAM: GOLD FUTURES VS SPOT (pp. 60-61)

Same 100-oz long-gold exposure for three months; 5% borrowing rate.

1. Futures route: post USD 4,950 initial margin $\rightarrow 4,950 \times 5\% \div 4 = \text{USD } 61.88$ of interest.
2. Spot route: pay $1,770 \times 100 = \text{USD } 177,000 \rightarrow 177,000 \times 5\% \div 4 = \text{USD } 2,212.50$ of interest.

Same exposure: USD 61.88 of carry against USD 2,212.50 – roughly 36× less.

Source: CFA Institute, Curriculum Vol 7, Learning Module 3, "Derivative Benefits", Exhibit 1, pp. 55–62 (printed); Example 4, pp. 60–61.

The curriculum names SIX risks, not four — implicit leverage, opacity, basis, liquidity, counterparty credit, and destabilisation of the system as a whole

The tools are powerful because they are leveraged; the same property makes an unmanaged position a route to systemic stress.

Risk (CFA Exhibit 3)	What it is	Where it bites
Speculative use (leverage)	Small margin controls large notional	Margin calls, forced unwind
Lack of transparency	Complexity hides the true exposure profile	Stakeholders misprice risk
Basis risk	Derivative underlying is not the hedged item	Residual P&L after hedging
Liquidity risk	Cash-flow TIMING mismatch vs hedged item	Margin call you cannot fund
Counterparty credit risk	OTC dealer may not perform on the contract	CCP, ISDA, collateral
Destabilisation / systemic	Leverage across the system amplifies stress	2008 – AIG, Lehman

BASIS VS LIQUIDITY – THE PAIR STUDENTS SWAP

Basis risk is a mismatch of VALUE: the derivative's underlying is similar to but not identical to the exposure (a WTI future against a Brent cargo, one MRR against another). Liquidity risk is a mismatch of TIMING: daily variation margin on a multi-year hedge against an annual coupon. One leaves residual P&L; the other can force you out of a hedge that was working.

¹ CFA LM 3 Ex 5 (p. 64): Procam controls USD 177,000 of gold risk on USD 61.88 of financing — a leverage ratio of 2,860.

Source: CFA Institute, Curriculum Vol 7, Learning Module 3, "Derivative Risks", Exhibit 3, p. 66 (printed); Example 5, pp. 63–64.

Issuers use derivatives to hedge commercial and financing exposures; investors use them to modify portfolio cash flows and replicate exposures

Issuers chase HEDGE ACCOUNTING — cash-flow, fair-value or net-investment — so the derivative's MTM lands in earnings alongside the item it hedges, not before it.

ISSUER USE CASES

- FX export hedge — Montau's KRW/EUR forward locks EUR revenue → CASH-FLOW hedge
- Commodity input hedge (airline buying jet-fuel swaps) → cash-flow hedge
- Floating debt → fixed via pay-fixed IRS → CASH-FLOW hedge
- Fixed debt → floating via receive-fixed IRS → FAIR-VALUE hedge
- FX forward or swap against a foreign subsidiary's equity → NET-INVESTMENT hedge

INVESTOR USE CASES

- Modify portfolio duration: receive-fixed IRS lengthens, pay-fixed shortens — no bond trading
- Covered call on existing long equity → extra yield, capped upside
- Synthetic index exposure via futures (low capital, high liquidity)
- Hedge FX exposure on foreign equities and bonds
- Express a credit view via CDS without owning the bond

Source: CFA Institute, Curriculum Vol 7, Learning Module 3, "Issuer Use" / "Investor Use of Derivatives", pp. 67–72 (printed).

SECTION 03

Arbitrage, replication & cost of carry

Derivative pricing rests on a single engine — the law of one price enforced by riskless arbitrage between spot and forward positions

If two positions deliver the same future cash flows, they must have the same price today. Any deviation is arbitrated away in seconds.

$$S_0 = F_0(T) / (1 + r)^T \quad \Leftrightarrow \quad F_0(T) = S_0 (1 + r)^T$$

where: S_0 = spot price today; $F_0(T)$ = forward price agreed today for delivery at T ; r = risk-free rate over horizon T .

ARBITRAGE

A trade that makes a RISKLESS positive profit with ZERO net investment. Requires simultaneous positions. In efficient markets arbitrage is fleeting — but it is what enforces the no-arbitrage condition.

REPLICATION

Reproduce a derivative's cash flows with a spot position plus risk-free borrowing or lending. Long forward $\equiv$ buy spot + borrow S_0 . If replication and derivative prices differ, arbitrage opens.

COST OF CARRY

Net cost of holding the underlying from 0 to T : opportunity cost r , PLUS costs C (storage, insurance), MINUS income I (dividends, coupons, convenience yield). F must cover the carry.

If a forward is mispriced relative to spot plus carry, a cash-and-carry trade locks in a riskless profit – the arbitrage that pins $F_0(T)$

Sell the rich leg, buy the cheap leg, carry to maturity. Every pricing formula in this deck is enforced by exactly this trade.

No-arb: $F_0(T) = S_0 (1+r)^T$
Observed $F > \text{No-arb} \Rightarrow \text{SELL forward} + \text{BUY spot} + \text{BORROW } S_0$

CFA LM 4 – QUESTION SET Q4, PRINTED p. 87

$S_0 = 100$; observed 3-month forward $F_0 = 102$; risk-free rate $r = 5\%$ p.a.; $T = 0.25\text{y}$. No dividend, no storage.

1. No-arbitrage forward: $F^*_0(T) = 100 \times (1.05)^{0.25} = 101.23$.
2. Observed $102 > 101.23 \Rightarrow$ the forward is RICH. Sell the forward, buy the spot, borrow 100 at 5% to fund it.
3. At $t = T$: deliver the asset you already hold and collect 102 under the forward.
4. Repay the loan: $100 \times (1.05)^{0.25} = 101.23$.
5. Profit = $102 - 101.23 = 0.77$, riskless, with zero net investment at $t = 0$.

Profit 0.77 pins the forward at 101.23. This trade IS the proof of $F_0(T) = S_0(1+r)^T$.

A long forward can be replicated by borrowing at r to buy the underlying at spot — the replication identity that gives $F_0(T) = S_0(1+r)^T$

Long forward $\equiv$ LONG asset, financed. Short forward $\equiv$ SHORT asset, proceeds lent. This identity is why a forward has $V_0 = 0$.

Position	$t = 0$ cash flows	$t = T$ cash flows	Net at T
LONG forward	0 ($V_0 = 0$)	Receive asset, pay $F_0(T)$	$S_{T,T} - F_0(T)$
Replicate: buy spot + borrow	Buy S at S_0 ; borrow S_0 at r	Asset = $S_{T,T}$; repay $S_0(1+r)^T$	$S_{T,T} - S_0(1+r)^T$
SHORT forward	0 ($V_0 = 0$)	Deliver asset, get $F_0(T)$	$F_0(T) - S_{T,T}$
Replicate: short spot + lend	Short S at S_0 ; lend S_0 at r	Cover at $S_{T,T}$; get $S_0(1+r)^T$	$S_0(1+r)^T - S_{T,T}$

THE IDENTITY THAT PRICES EVERY FORWARD

For the two long rows to be indifferent, $S_{T,T} - F_0(T)$ must equal $S_{T,T} - S_0(1+r)^T$, which forces $F_0(T) = S_0(1+r)^T$. Note what falls out: $S_{T,T}$ cancels, so the identity holds for ANY future spot price — that is what makes it riskless. A negative r flips the sign and puts F_0 below S_0 . Add carry and the same identity prices every asset class in this deck.

Cost of carry adjusts the replication formula: costs of ownership are ADDED and income benefits are SUBTRACTED before compounding at r

Dividends, coupons and convenience yields are BENEFITS the spot holder gets and the forward buyer does not. Storage, insurance and spoilage are added COSTS.

$$\begin{aligned} \text{Discrete: } F_0(T) &= [S_0 - PV(I) + PV(C)] (1 + r)^T \\ \text{Continuous: } F_0(T) &= S_0 \cdot e^{(r + c - i) T} \end{aligned}$$

where: I, C = income and cost in present-value terms; i, c = income and cost as continuous rates; r = risk-free rate. Costs carry a PLUS, income a MINUS.

Asset class	Benefit (I, i)	Cost (C, c)	Forward vs spot
Non-dividend equity	None	None	$F > S$ (carry = r only)
Dividend stock / index	Dividend I or yield i	None	$F > S$ iff $r > i$
Coupon bond (at par)	Coupon income I	None	$F < S$ iff coupon $> r$
FX, quoted f/d	Base-ccy rate r_d	Price-ccy rate r_f	$F > S$ iff $r_f > r_d$
Physical commodity	Convenience yield	Storage, insurance	$F > S$ unless yield wins

Sign check: if F came out ABOVE spot on a high-dividend stock, you added the income instead of subtracting it.

¹ Contango ($F > S$) / backwardation ($F < S$) are market terms not used in Vol 7. A forward far below the no-arb price signals a convenience yield (p. 94).

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, "Costs and Benefits of Owning the Underlying", Exhibits 8 and 10, pp. 88–96 (printed).

SECTION 04

Forwards & futures

PRICE vs VALUE: $F_0(T)$ is set at inception so that $V_0(T) = 0$ for both sides — the price never changes again, the value changes every day

No cash changes hands at $t = 0$ because the contract price already embeds the carry. Value only becomes non-zero once S , r or time move.

$$\begin{aligned} F_0(T) &= S_0 (1 + r)^T \\ V_0(T) &= 0 \quad (\text{to both long and short}) \end{aligned}$$

where: $F_0(T)$ is the PRICE — fixed for the life of the contract. $V_t(T)$ is the VALUE — marked to market continuously.

CFA LM 5 — VFO SELLS BIOMIAN FORWARD (PRINTED pp. 104-107)

VFO sells 1,000 Biomian shares forward to an intermediary. $S_0 = \text{INR } 295$; $T = 0.5\text{y}$; no dividend; $r = 4\%$.

1. $F_0(T) = 295 \times (1.04)^{0.5}$; $(1.04)^{0.5} = 1.01980$
2. $F_0(T) = 295 \times 1.01980 = \text{INR } 300.84$ per share
3. $V_0(T) = 0$ for BOTH sides — the 4% is already inside the 300.84
4. At maturity, $S_T = 287 \Rightarrow \text{VFO (short) gains } 13.84$; $S_T = 312 \Rightarrow \text{VFO loses } 11.16$

$F_0(T) = \text{INR } 300.84$. Contract size $1,000 \times 300.84 = \text{INR } 300,840$.

After inception, the MTM of a long forward is current spot minus the PV of the original forward price — and a higher r shrinks the short's gain

Intuition: a forward is a PROMISE to pay F_0 at T . That promise re-values whenever the PV of F_0 drifts away from the current spot.

$$\begin{aligned}V_t(T) \text{ long} &= S_t - F_0(T)(1+r)^{-(T-t)} \\V_t(T) \text{ short} &= F_0(T)(1+r)^{-(T-t)} - S_t\end{aligned}$$

where: S_t = spot price now; $T - t$ = remaining life; r = current risk-free rate. Note $F_0(T)$ is unchanged — only its PV moves.

CFA LM 5 EXAMPLE 3 — VFO MTM AT $t = 0.25$ (PRINTED p. 108)

VFO is SHORT: $F_0 = \text{INR } 300.84$, $r = 4\%$, $T = 0.5\text{y}$. At $t = 0.25$ spot has fallen to $S_t = \text{INR } 285$.

1. PV of F_0 at t : $300.84 \times (1.04)^{-0.25} = 300.84 \times 0.99024 = \text{INR } 297.90$
2. Short MTM: $V_t = F_0(1+r)^{-(T-t)} - S_t = 297.90 - 285 = \text{INR } 12.90$ per share
3. Total: $1,000 \times 12.90 = \text{INR } 12,900$ gain to VFO
4. Now double r to 8% : $\text{PV}(F_0) = 300.84 \times (1.08)^{-0.25} = 295.11 \rightarrow \text{MTM } 10.11$, a fall of 2.79

Short gains 12.90/share at $r = 4\%$, only 10.11/share at $r = 8\%$.

Equity forwards with discrete dividends price at $F_0(T) = [S_0 - PV(I)](1+r)^T$ – the dividend accrues to the spot holder, not the forward buyer

Rule: subtract the PV of every dividend paid BEFORE maturity from S_0 before compounding at r . Dividends paid AFTER T are irrelevant.

$$F_0(T) = [S_0 - PV_0(I)] \cdot (1 + r)^T$$

where: $PV_0(I)$ = present value at $t = 0$ of every income payment (dividend, coupon) received over the interval $[0, T]$.

CFA LM 4 EXAMPLE 4 – HIGHEST / UNILEVER, $T = 0.5y$ (PRINTED p. 90)

Highest delivers 1,000 UL shares in six months. S_0 = EUR 50. UL pays a EUR 0.30 dividend at $t = 0.25$ and again at $t = 0.5$. $r = 5\%$.

1. $PV(I) = 0.30 \times (1.05)^{-0.25} + 0.30 \times (1.05)^{-0.5} = 0.2964 + 0.2928 = \text{EUR } 0.5892$
2. $F_0(T) = (50 - 0.5892) \times (1.05)^{0.5} = 49.4108 \times 1.02470 = \text{EUR } 50.6310$
3. Contract value = $1,000 \times 50.6310 = \text{EUR } 50,631.00$
4. Without dividends: $50 \times (1.05)^{0.5} = 51.23$. Carry over spot falls from 1.23 to 0.63.

$F_0 = \text{EUR } 50.6310$ – dividends cut the forward premium from EUR 1.23 to EUR 0.63.

For an equity index, dividends are modelled as a continuous yield – the forward compounds spot at the NET carry, r minus i , not at r alone

For indexes, dividends are modelled as a CONTINUOUS yield i because the constituents smear payment dates across the whole year.

$$F_0(T) = S_0 \cdot e^{(r + c - i)T}$$

for an index $c = 0 \Rightarrow F_0 = S_0 \cdot e^{(r - i)T}$

where: i = dividend yield (continuous); c = carry cost rate (continuous); r = risk-free rate (continuous).

CFA LM 4 EXAMPLE 5 – VFO NIFTY 50, 3 MONTHS (PRINTED p. 91)

Spot NIFTY 50 = INR 15,200. Index dividend yield $i = 2.2\%$. INR risk-free $r = 4.0\%$. $T = 0.25y$.

1. Net carry rate = $r - i = 4.0\% - 2.2\% = 1.8\%$
2. Exponent = $0.018 \times 0.25 = 0.0045$
3. $e^{0.0045} = 1.004510$
4. $F_0 = 15,200 \times 1.004510 = \text{INR } 15,268.55$

$F_0 = \text{INR } 15,268.55$. If i exceeded r the forward would sit BELOW spot.

FX forwards price off covered interest-rate parity — the forward is spot grown at the interest-rate DIFFERENTIAL, never at the level of rates

Borrow domestic, convert at spot, lend foreign, lock the rate forward. Two riskless strategies must return the same — that equality forces F .

$$F_{\{0, f/d\}}(T) = S_{\{0, f/d\}} \cdot e^{(r_f - r_d) T}$$

where: f = PRICE currency (numerator); d = BASE currency (denominator). S = units of f per one unit of d . Forward quoted the same way as spot.

CFA LM 4 EXAMPLE 6 — AUD/USD SIX-MONTH FORWARD (PRINTED pp. 91-92)

Spot $S_0 = 1.3335$ AUD/USD (AUD is the price currency f , USD the base currency d). 6m AUD $r_f = 0.05\%$; 6m USD $r_d = 0.20\%$. $T = 0.5y$.

1. Differential $(r_f - r_d) = 0.0005 - 0.0020 = -0.0015$ p.a.
2. Exponent $= -0.0015 \times 0.5 = -0.00075$; $e^{-0.00075} = 0.999250$
3. $F_0 = 1.3335 \times 0.999250 = 1.3325$ AUD/USD
4. Check by replication: borrow USD 1,000 $\rightarrow$ AUD 1,333.50 $\rightarrow$ lend $\rightarrow$ AUD 1,333.83 vs USD 1,001 repaid $\Rightarrow 1.3325$

$F_0 = 1.3325$. Fewer AUD buy a USD later $\Rightarrow$ AUD at a forward PREMIUM — the LOWER-rate currency.

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, Example 6 (AUD/USD replication) and Exhibit 9, pp. 91–93 (printed).

Montau: the higher-yielding KRW trades at a forward DISCOUNT — and a three-legged money-market trade proves the forward rate to the last won

If F deviates from $S \cdot e^{(r_f - r_d)T}$, a three-legged trade — spot plus two money-market legs — locks in riskless profit. So it does not deviate.

$$F_{\{KRW/EUR\}}(T) = S_{\{KRW/EUR\}} \cdot e^{(r_{KRW} - r_{EUR})T}$$

CFA LM 4 EXAMPLE 7 — MONTAU AG, KRW 650m RECEIVABLE IN 75 DAYS

Spot $S_0 = 1,300$ KRW/EUR. $r_{KRW} = 0.75\%$ (price currency). $r_{EUR} = -0.25\%$ (base currency). $T = 75/365$ year.

1. Differential $(r_{KRW} - r_{EUR}) = 0.0075 - (-0.0025) = 0.0100$
2. Exponent $0.0100 \times (75/365) = 0.002055 \Rightarrow F_0 = 1,300 \times e^{0.002055} = 1,302.67$ KRW/EUR
3. Proof, EUR leg: KRW 650m $\div 1,300 =$ EUR 500,000 grown at $-0.25\% \rightarrow$ EUR 499,743.22
4. Proof, KRW leg: KRW 650m grown at $0.75\% \rightarrow$ KRW 651,002,484.59
5. Implied forward $= 651,002,484.59 / 499,743.22 = 1,302.67$ ✓

$F_0 = 1,302.67$ KRW/EUR. More KRW per EUR later $\Rightarrow$ KRW at a forward DISCOUNT (higher rate).

Futures are forwards with daily mark-to-market — cumulative P&L matches a forward's, but the cash-flow timing is completely different

Each evening the MTM is settled in cash and the contract value RESETS to zero. Cumulative MTM through $T \approx$ the forward's single payoff at T .

$$\text{Daily MTM (long)} = [f_t(T) - f_{t-1}(T)] \times \text{contract size}$$

where: $f_t(T)$ = futures settlement price on day t . After settlement the contract is re-struck at $f_t(T)$ and its value is zero again.

CFA LM 6 QUESTION SET Q2 — SHORT CRUDE OIL, DAY T-5 (pp. 139-140)

Investor is SHORT 1,000 barrels of crude entered at f_0 = USD 69.00. Day T-6 settled at 66.02; day T-5 settled at 68.01.

1. Daily futures price change: $68.01 - 66.02 = +1.99$ per barrel
2. MTM to the LONG = $+1.99 \times 1,000 = +\text{USD } 1,990$
3. MTM to the SHORT = $-1.99 \times 1,000 = -\text{USD } 1,990$, debited from margin that evening
4. The price ROSE, so the short loses. Sign follows the position, not the direction of the market.

Short loses USD 1,990 on day T-5 — realized that evening, not at maturity.

Source: CFA Institute, Curriculum Vol 7, Learning Module 6, Question Set "Valuation and Pricing of Futures Contracts", Q2, pp. 139–140 (printed).

A margin call triggers when the balance falls below MAINTENANCE margin — but the top-up must restore the account to INITIAL margin, not to maintenance

Initial margin is the entry ticket. Maintenance margin is the trigger level. The gap between them is how far you can lose before the phone rings.

$$f_0(T) = [S_0 + PV_0(C)] (1 + r)^T$$

Margin call when Balance < Maintenance $\Rightarrow$ restore to INITIAL

CFA LM 6 PRACTICE PROBLEM 1 — CME COPPER, LONG POSITION

Contract 25,000 lb. $S_0 = \text{USD } 4.25/\text{lb}$. Storage USD 10/contract payable at T. $r = 1.875\%$. $T = 1/12$. Initial margin 10,000; maintenance 6,000.

1. PV of storage: $PV_0(C) = 10 \times (1.01875)^{-(1/12)} = \text{USD } 9.98$ per contract
2. $f_0 = [(4.25 \times 25,000) + 9.98] \times (1.01875)^{(1/12)} = \text{USD } 106,425 \approx \text{USD } 4.257/\text{lb}$
3. Call triggers after a loss of $10,000 - 6,000 = \text{USD } 4,000$
4. $\text{USD } 4,000 / 25,000 \text{ lb} = \text{USD } 0.16/\text{lb} \Rightarrow \text{call at } 4.257 - 0.16 = \text{USD } 4.10/\text{lb} (= 102,425/\text{contract})$
5. The CTA must then deposit enough to return the account to USD 10,000 — the INITIAL margin.

Margin call below USD 4.10/lb; the top-up restores INITIAL margin, not maintenance.

¹ Basis ($S - f$) converges to zero at maturity: in the CFA crude-oil table, spot and futures both settle at USD 70.03 on day T (p. 139).

Source: CFA Institute, Curriculum Vol 7, Learning Module 6, Practice Problem 1 (CME copper), pp. 146 and 148 (printed); margin rule at p. 135.

Forward and futures prices are equal if rates are constant or uncorrelated with price — correlation and, separately, convexity are what pull them apart

Short-dated contracts: the gap is a rounding error. Multi-year rate contracts (SOFR futures vs IRS): it is a basis dealers quote and charge for.

MECHANISM 1 — MTM CORRELATION

- EQUAL if r is constant, OR if r is uncorrelated with the futures price.
- Futures > forward when f is POSITIVELY correlated with r — daily gains get reinvested at higher rates, losses financed at lower ones.
- Futures < forward when f is NEGATIVELY correlated with r — e.g. bond futures, where price falls as rates rise.
- Rule of thumb: the more desirable contract carries the higher price.

MECHANISM 2 — CONVEXITY BIAS

- Rate FUTURES settle linearly — a fixed BPV per basis point. An FRA settles on the PRESENT VALUE of the final cash flow.
- CFA Ex 5: $\pm 20\text{bp}$ on USD 1m of 3m MRR gives futures ± 500 but FRA $+497.50 / -497.01$. Not symmetric.
- That asymmetry is positive convexity and it favours the FORWARD holder. So the futures-implied RATE prints ABOVE the FRA/swap rate — subtract a convexity adjustment to get the forward rate.
- The bias grows with tenor and rate volatility. Central clearing narrows the cash-flow gap, not this one.

Source: CFA Institute, Curriculum Vol 7, Learning Module 6, "Forward and Futures Price Differences" and "Interest Rate Forward and Futures Price Differences", pp. 141–143 (printed).

SECTION 05

Swaps

A swap is a series of firm commitments — economically a strip of forwards, but with ONE constant fixed rate instead of period-specific forward rates

One contract with one fixed rate replaces N FRAs with N different fixed rates — fewer tickets, one credit line, the same aggregate exposure.

$$\text{Settlement (pay-fixed)} = (\text{MRR} - s_N) \times \text{Notional} \times \text{Period}$$

where: s_N = the fixed swap rate for an N-period swap. MRR sets at the START of each period and is settled at the END of it.

STRIP OF FRAs

Each FRA carries its OWN fixed rate — the implied forward rate for that period. N contracts, N rates, N credit exposures. And an FRA settles ONCE, at the START of its interest period, on a discounted basis.

INTEREST-RATE SWAP

ONE contract and ONE constant fixed rate s_N , set so that $PV(\text{fixed}) = PV(\text{floating})$ at $t = 0$. Same aggregate exposure, one ticket — and each settlement falls at the END of its period, undiscounted.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, "Swaps vs. Forwards", Exhibits 1–2, pp. 152–154 (printed).

The par swap rate is the ONE fixed rate that makes PV(expected floating) equal PV(fixed) at $t = 0$ — an internal rate of return on the forward curve

s_N is chosen precisely so the swap is worth nothing to either side on day one. That is what "priced at par" means.

$$\sum IFR_i / (1+z_i)^i = s_N \times \sum 1 / (1+z_i)^i \quad (i = 1 \dots N)$$

where: IFR_i = implied forward rate for period i (derived from the zero curve); z_i = zero rate for maturity i ; s_N = par swap rate.

THE INTUITION IN THREE LINES

LEFT = PV(floating): each period's implied forward rate discounted at its matching zero rate. RIGHT = PV(fixed): the level rate s_N multiplied by the sum of discount factors. Set them equal, solve for s_N — and by construction $V_0 = 0$.

Geometrically, s_N is a horizontal line drawn through the forward curve so that the PV-weighted area above equals the PV-weighted area below. An upward-sloping curve pushes s_N above the first forward rate; a downward-sloping curve pulls it below.

Crucially, individual exchanges are NOT worth zero. The early ones are negative to the fixed payer and the late ones positive. Only the SUM is zero.

Worked: a 3-year IRS prices at $s_3 = 3.9641\%$ on a zero curve of 2.3960% / 3.4197% / 4.0005%

Step 1: bootstrap the implied forwards from the zero curve. Step 2: plug into the par-swap equation. Step 3: divide.

Period	Zero rate z_i	Implied forward IFR_i	Derivation
1	2.3960%	2.3960%	$IFR_{\{0,1\}} = z_1$
2	3.4197%	4.4536%	$(1.034197)^2 / (1.02396) - 1$
3	4.0005%	5.1719%	$(1.040005)^3 / (1.034197)^2 - 1$

SOLVE FOR s_3 (CFA PRINTED p. 155)

1. Set $PV(\text{float}) = PV(\text{fixed})$: $\sum IFR_i / (1+z_i)^i = s_3 \times \sum 1 / (1+z_i)^i$
2. $PV(\text{float}) = 2.3960 / 1.02396 + 4.4536 / (1.034197)^2 + 5.1719 / (1.040005)^3 = 0.111017$
3. Discount factors: $0.976601 + 0.934962 + 0.888982 = 2.800545$
4. $s_3 = 0.111017 / 2.800545 = 3.9641\%$

$s_3 = 3.9641\%$ – between the 2.396% front IFR and the 5.172% back IFR, as an upward curve requires.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, Example 1 (Swaps as a Combination of Forwards), pp. 153–155 (printed).

$V_0(\text{swap}) = 0$ by construction; on any settlement date the swap is worth the current settlement plus the PV of every remaining exchange

At $t = 0$ $PV(\text{fixed}) = PV(\text{floating})$ by design. As time passes and rates move, the legs drift apart — and that gap IS the swap MTM.

$$V_t(\text{swap}) = \text{current period settlement} + PV_t(\text{all remaining fixed-vs-floating exchanges})$$

where: Settlement = $(MRR - s_N) \times \text{Notional} \times \text{Period}$ for the fixed-rate PAYER; the sign reverses for the receiver.

CFA LM 7 EXAMPLE 3 — ESTERR, CAD 250m PAY-FIXED AT 2.05%

Esterr pays fixed 2.05% quarterly and receives 3m CAD MRR on CAD 250m. Period = 0.25. The IFR curve rises ~0.24% per quarter from 0.50%.

1. Period 1: $(0.50\% - 2.05\%) \times 250\text{m} \times 0.25 = -\text{CAD } 968,750$ — Esterr pays.
2. Period 2: $(0.74\% - 2.05\%) \times 250\text{m} \times 0.25 = -\text{CAD } 818,750$ — Esterr still pays.
3. Esterr keeps paying through period 7 ($1.94\% < 2.05\%$) and starts RECEIVING in period 8 ($2.18\% > 2.05\%$).
4. Roll forward with the curve unchanged: the loss-making quarters fall away, so $PV(\text{floating received}) > PV(\text{fixed paid}) \Rightarrow \text{MTM gain}$.

$V_0 = 0$. Esterr pays through period 7, receives from period 8 — MTM gain as the payers roll off.

An interest-rate swap is a pair of bonds — pay-fixed $\equiv$ long FRN + short fixed bond; receive-fixed $\equiv$ long fixed bond + short FRN

The principals cancel at inception and again at maturity. Only the coupon legs survive — so a swap is two bond coupons traded with a partner.

Swap side	Equivalent bond portfolio	Effective duration	When rates RISE
Pay-fixed / Receive-floating	LONG FRN + SHORT fixed bond	Negative	MTM GAIN
Receive-fixed / Pay-floating	LONG fixed bond + SHORT FRN	Positive	MTM LOSS

WHY THE PRINCIPALS CANCEL

Take the RECEIVE-fixed side. At $t = 0$ the manager buys a fixed-rate bond for N ($-N$ cash) and funds it by issuing an FRN at MRR ($+N$ cash): net zero. At maturity the bond repays N ($+N$) and the FRN is redeemed at N ($-N$): net zero again. Only the coupon legs survive — receive fixed, pay MRR — which is exactly a receive-fixed swap. Flip every sign and you have pay-fixed. This is why a swap carries the DIFFERENCE in duration and DV01 between the two bonds, not the level of either, and why an FRN contributes almost no duration: it reprices to par at every reset.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, Exhibit 4 "Using Swaps to Manage Fixed-Income Exposure", p. 157, and p. 162 (printed).

Post-inception, a rise in expected forward rates raises PV(floating) — an MTM gain for the fixed-rate payer and a loss for the fixed-rate receiver

Receive-fixed IS a long fixed-rate bond; pay-fixed IS a short one. Every row below follows from that one equivalence.

Event post-inception	Effect on PV(float)	Effect on PV(fixed)	Pay-fixed MTM	Receive-fixed MTM
Forward curve shifts UP parallel	Up	Down (higher disc)	GAIN	LOSS
Forward curve shifts DOWN parallel	Down	Up (lower disc)	LOSS	GAIN
Time passes, upward curve unchanged	Float > Fixed	—	GAIN	LOSS
Time passes, downward curve unchanged	Float < Fixed	—	LOSS	GAIN

DURATION INTUITION

A receive-fixed swap behaves like a long fixed-rate bond — positive duration. A manager who wants MORE portfolio duration without buying bonds receives fixed; one who wants LESS pays fixed. Rows 3 and 4 are the ones students miss: on an unchanged upward-sloping curve, the mere passage of time hands the fixed payer a gain, because the cheap early periods have already been paid away.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, "Swap Values and Prices" and Question Set, pp. 159–165 (printed).

Worked: Fyleton receives fixed 2.38% on GBP 200m — time on an upward curve costs it, a downward shift in the curve pays it, exactly like a long bond

Receive-fixed is a long-bond proxy. Rates fall, bonds rally, the swap gains — same directional exposure, far cheaper to put on.

$$\text{Settlement (receive-fixed)} = (s_N - \text{MRR}) \times \text{Notional} \times \text{Period}$$

CFA LM 7 EXAMPLE 4 — FYLETON, RECEIVE-FIXED 5y GBP 200m

Fyleton receives fixed 2.38% semi-annually and pays 6m GBP MRR on GBP 200m, to lengthen portfolio duration. Initial MRR sets at 0.71%. Period = 0.5. Forward curve slopes UP.

1. First settlement at $t = 0.5$: $(2.38\% - 0.71\%) \times 200\text{m} \times 0.5 = +\text{GBP } 1,670,000$ received.
2. Case A — time passes, curve unchanged: the MRRs Fyleton pays climb past 2.38%, so $\text{PV}(\text{floating paid}) > \text{PV}(\text{fixed received}) \Rightarrow \text{MTM LOSS}$.
3. Case B — the forward curve shifts DOWN: $\text{PV}(\text{floating paid})$ falls while the fixed leg is unchanged $\Rightarrow \text{MTM GAIN}$.
4. Same signature as a long fixed-rate bond: it bleeds as an upward curve rolls through, and rallies when the curve drops.

First exchange +GBP 1.67m. Time on an upward curve = LOSS; a downward curve shift = GAIN.

Deck 17 — Derivatives I: Forwards, Futures & Swaps

■ Firm commitments (forwards, futures, swaps): symmetric payoff, $V_0 = 0$. Contingent claims (options, CDS): asymmetric, and cost a premium. The long is the BUYER.

■ $F_0(T) = S_0(1+r)^T$ is a PRICE, not a forecast. With carry: $F_0 = [S_0 - PV(I) + PV(C)](1+r)^T$ — costs ADD, income SUBTRACTS.

■ FX parity: $F_{f/d} = S_{f/d} \cdot e^{(r_f - r_d)T}$. The LOWER-rate currency trades at a forward PREMIUM. AUD/USD 1.3335 $\rightarrow$ 1.3325; Montau KRW/EUR 1,300 $\rightarrow$ 1,302.67.

■ Par swap rate solves $\sum IFR_i / (1+z_i)^i = s_N \times \sum 1 / (1+z_i)^i$. Three-year example: $s_3 = 3.9641\%$ on a 2.396 / 3.4197 / 4.0005% zero curve. $V_0 = 0$ in aggregate, not per period.

■ OTC = customized and bilateral under an ISDA; ETD = standardized, novated to a CCP, margined daily. Post-2008 clearing of standardized OTC has narrowed the gap.

■ Price is fixed at inception so $V_0 = 0$; value moves later. Long MTM: $V_t = S_t - F_0(T)(1+r)^{-(T-t)}$. Biomian: $F_0 = 300.84$, short gains 12.90 at $t = 0.25$ (10.11 if r doubles to 8%).

■ Futures = forwards plus daily MTM; a call triggers at MAINTENANCE and restores to INITIAL. Prices split on MTM correlation and, for rate contracts, convexity.

■ IRS $\equiv$ a pair of bonds: pay-fixed $\equiv$ long FRN + short fixed bond (negative duration); receive-fixed $\equiv$ long fixed bond + short FRN (positive duration). Rates up $\Rightarrow$ pay-fixed gains.